Micropolis Eduverse Progress Report

Educational Open Source SimCity,
delivered over the World Wide Web.

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Micropolis Eduverse Goals

- Develop a fun web based educational game, based on the original SimCity source code.
- Focused on constructionist education.
- Built out of free open source software.
- Runs on the server, plays in the browser.
- Supports collaborative, democratic, multi-user play.

Achievements So Far

- Developed an open source, internationalized, web based version of Micropolis.
- Client based on OpenLaszlo/Flash on the client.
- Server based on TurboGears/Python/C++.
- Also developed a cross platform, desktop version of Micropolis, based on GTK/Python.
- Renovated, refactored and documented the code.
- Established a solid groundwork and open framework for developling an educational, collaborative, multi-player game.

Open Source Software

- Micropolis is built with a carefully chosen, well proven, widely supported set of open source, cross platform tools and libraries.

- Micropolis itself and its TileEngine module are open source (GPL3). The client runs on the proprietary Adobe Flash player, which is installed in about 98% of all web browsers. The rest of the software it depends on is free open source.

- Linux. Apache. MySQL. Python. Cairo. Pango. GTK. GCC. SWIG. PyCairo. PyGTK. TurboGears. Genshi. SQLAlchemy. CherryPy. PyAMF. AMFast. Pie menus. OpenLaszlo. Red5. Java. Tomcat. WikiMedia. PHP.

Internationalized UI

- All text comes from Unicode XML localization files, so the text can be translated to support different languages and countries.
- Currently supports English and Dutch, but other languages are easy to add, given translation resources.
- Web based tool for conveniently reviewing and editing translations.
- This is a very important feature, since we're initially delivering to the Dutch educational system, and we intend to support many other languages in the future.

Deployment

- Server: Linux, Apache, MySQL, Python, TurboGears, C++
- Portable, cross platform C++ and Python code.
  - Linux and Unix
  - Mac OS/X
  - Windows (not currently maintained but viable)
- Deploys on standard Fedora Linux server, i.e. Amazon Elastic Computing Server, etc.
- Client: OpenLaszlo, XML, JavaScript, AMF

Multi Player Collaboration

- Don Hopkins designed, developed and shipped a multi-player X11 version of SimCity in 1993 for DUX software, demonstrated in Amsterdam at INTERCHI '93.
- Micropolis has been systematically redesigned with multi-player collaboration and constructionist education in mind.
- Currently laying the groundwork for multi player and online community support, while getting the single player game up and running first.
- User accounts. Saved cities. Shared cities. Chat. Robots and Avatars. Basic MediaWiki integration. Client/server session, messaging and simulation engine plumbing.

Next Steps

- Shared city library. Ratings. Tagging. Journals. Discussions. Linking. History browsing. Sharing. Voting. Profiles. Friends.

- Text, voice and video chat. In-game chat with avatars on the map. Integrated video player for help, tutorials and lessons. Integrated html frame for synchronized web browsing.

- MediaWiki integration, to support creative writing by students and courseware development by teachers.

- Facebook integration. Having your friends vote on issues to fund your grant proposals. Write a proposal to convince five of your friends to fund your proposal to build an airport.

- PayPal integration. Donations, subscriptions and micro-

Constructionist Education

- The original SimCity was an exemplary constructionist educational game, which has been successfully used in many classrooms, but it was a closed commercial product, not specifically designed to support education.
- Since Micropolis is open source SimCity, we can open up, renovate, visualize, document and extend it in wonderful ways, to specifically support constructionist education, social networking and collaborative democratic play.
- [Alan Kay Quote]

Supporting Education

- Encourage creative writing by blogging, twittering, chatting, writing newspaper articles about events and issues in the city.
- Teach democratic processes. Writing proposals. Campaigning for support. Discussing and debating issues. Voting on bills.
- Publish wiki pages with annotated maps and timelines of events and writing about the cities.
- Open up the simulator to enable plug-in programmable zones, tools, robots, avatars, events, policies, courseware, etc.
- Teach older kids to program Python and C++. Implement a visual behavior programming language (like Robot Odyssey